

Teleporting ridden entity in unloaded chunks does not cause chunks to load for riding player

MC-44514 Networking

Status	Resolved / Fixed
Affected	Minecraft 14w02b, Minecraft 14w02c, Minecraft 14w03a, Minecraft 14w05a, Minecraft 14w07a, Minecraft 14w08a (+71 more)
Fixed in	23w12a
Reported	2014-01-10

STEPS TO REPRODUCE

1. Launch Minecraft and create a new Creative world with default settings.
2. Place a boat.
3. Enter the boat.
4. Press F3 to open the debug screen.
5. Run `/tp @e[type=minecraft:boat,sort=nearest] ~ ~ ~5000``.
6. Run `/execute if block ~ ~ ~ air run say @s``.
7. Observe that the player is teleported client-side but remains visually stuck near the original location, the screen jitters, and the F3 coordinates show only a partial movement instead of the complete teleport distance; the chunks around the boat do not load, and the command indicates that the server-side position has not moved to the destination.

OBSERVED BEHAVIOUR

Actual behavior

When a ridden entity is teleported into an unloaded area, the destination chunks are not loaded for the riding player. The ridden entity is moved server-side, but the player is not synchronously moved with it because the normal passenger update does not occur while the destination is unloaded.

As a result, the client and server become desynchronized:

- The client may display the player at or near the teleported destination, while the server still considers the player to be at the original location.
- Depending on the version, the player may appear stuck, experience visual jitter, or be dismounted client-side.
- The F3 coordinates reflect the client-side position and may not match the player's server-side position.
- Block interactions and commands that depend on the player's position are generally evaluated using the server-side position and may therefore be rejected or behave inconsistently.
- The destination chunks are not loaded around the player as part of the teleport.

Executing `/ride @s dismount`` or relogging causes the server to synchronize the player's position with the ridden entity and load the chunks around the destination.

EXPECTED BEHAVIOUR

When a ridden entity is teleported into otherwise unloaded chunks, the chunks around the destination should load for the riding player, and the player should remain mounted and be moved there correctly on both the client and server. The player's coordinates, appearance, and interactions should remain synchronized, allowing commands, block breaking, and block placement to work normally at the destination.

ENVIRONMENT

Minecraft Java Edition Versions: 14w02b, 14w02c, 14w03a, 14w05a, 14w07a, 14w08a, 14w10a, 14w17a, 14w19a, 14w25a, 14w28a, 14w28b, 14w30a, 14w31a, 14w32b, 14w33a, 14w34d, 1.8-pre1, 1.8-pre3, 1.8, 1.8.3, 15w49b, 16w33a, 1.11.2, 1.12.1, 1.12.2, 1.13-pre8, 1.13, 18w30b, 18w31a, 18w32a, 18w33a, 1.13.1-pre1, 1.13.1, 1.13.2, 18w43b, 18w45a, 18w46a, 18w48a, 18w48b, 18w49a, 19w04b, 19w07a, 19w12b, 19w13b, 19w14a, 19w14b, 1.14.3, 1.14.4, 1.15.2, 20w08a, 20w12a, 20w14a, 20w19a, 20w21a, 20w22a, 1.16 Pre-release 3, 1.16 Pre-release

5, 1.16 Pre-release 7, 1.16 Release Candidate 1, 1.16, 1.16.1, 20w29a, 1.16.2, 1.16.4, 20w51a, 21w03a, 1.16.5, 21w05a, 21w05b, 21w06a, 1.17, 1.18, 1.18.1, 1.19, 1.19.2, 23w04a (includes snapshots)

ORIGINAL DESCRIPTION

The bug

Teleporting an entity which is ridden by a player into unloaded chunks teleports the player (client-side) but does not load the chunks.

In current versions, the player becomes stuck visually at their current position and their screen jitters, but the coordinates on the debug screen show that they moved some (but not the complete) distance.

In 1.18, the boat teleports but the player desyncs and cannot interact with anything that is not near the location where they typed the command.

As of 23w04a, teleporting the boat into unloaded chunks dismounts the player client-side. If the player tries breaking or placing blocks after running the command, these actions will generally be invalidated by the server, although in some circumstances they seem to work even though the server thinks the player is far away. The F3 coordinates seem to consistently reflect where the player appears to be (on the client). If the player runs `/ride @s dismount` or `/relog`, they will be teleported to the boat and will load the chunks around it.

How to reproduce

- Place a boat and enter it
- Use the following command

```
/tp @e[type=minecraft:boat,sort=nearest] ~ ~ ~5000
```

- Enable the debug information

→ You will see that you were teleported (client-side), but using for example commands which print the block coordinates like `/execute if block ~ ~ ~ air run say @s` show that server-side you have not moved

The screenshots show a mounted horse before teleporting at 0,1000; the "void" after teleporting the horse; and the mounted horse at 0,5000 right after relogging.

Code analysis

Based on 1.11.2 decompiled using MCP 9.35 rc1

The reason is the same one as for

MC-92916

Resolved

and MC-108469: The method `net.minecraft.world.World.updateEntityWithOptionalForce(Entity, boolean)` does not update entities which are not in loaded chunks server-side. The problem here is that the riding player is not instantly moved with the ridden entity but instead when the method `Entity.updateRidden()` is called, which in this case never happens because the area around the ridden entity is not loaded.

MANUAL RESULT

Version used	
Reproduced?	YES / NO / PARTIAL
Crash file	
Notes	
Tested by / date	

Live issue: <https://bugs.mojang.com/browse/MC/issues/MC-44514>